

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE loans (
  ->     loan_id INT PRIMARY KEY,
  ->     member_id INT,
  ->     book_id INT,
  ->     loan_date DATE,
  ->     FOREIGN KEY (member_id) REFERENCES members(member_id),
  ->     FOREIGN KEY (book_id) REFERENCES books(book_id)
  -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| loans                  |
| members                |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| loans                  |
| members                |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO loans
-> (loan_id, member_id, book_id, loan_date)
-> VALUES
-> (1,1,1,'2026-07-01'),
-> (2,3,3,'2026-07-03'),
-> (3,1,3,'2026-07-05'),
-> (4,3,1,'2026-07-07');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
|      1 |         1 |        1 | 2026-07-01 |
|      2 |         3 |        3 | 2026-07-03 |
|      3 |         1 |        3 | 2026-07-05 |
|      4 |         3 |        1 | 2026-07-07 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT
-> m.full_name AS Member_Name,
-> b.title AS Book_Title
-> FROM loans l
-> INNER JOIN members m
-> ON l.member_id=m.member_id
-> INNER JOIN books b
-> ON l.book_id=b.book_id;
```

Member_Name	Book_Title
Anil Kumar	The Alchemist
Anil Kumar	Atomic Habits
Ravi Verma	Atomic Habits
Ravi Verma	The Alchemist

4 rows in set (0.00 sec)

```
mysql> SELECT
-> published_year,
-> COUNT(book_id) AS Total_Books
-> FROM books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1988	1
2018	1

2 rows in set (0.02 sec)

```
mysql> CREATE TABLE donation_history(  
->     donation_id INT PRIMARY KEY,  
->     book_id INT,  
->     donor_name VARCHAR(100),  
->     donation_date DATE,  
->     FOREIGN KEY(book_id) REFERENCES books(book_id)  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO books  
-> (book_id,title,isbn,published_year)  
-> VALUES  
-> (4,'Python Programming',  
-> '9781234567890',  
-> 2025);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO donation_history  
-> (donation_id,book_id,donor_name,donation_date)  
-> VALUES  
-> (1,4,'Prem Kumar',CURDATE());  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM books;  
+-----+-----+-----+-----+  
| book_id | title           | isbn           | published_year |  
+-----+-----+-----+-----+  
| 1       | The Alchemist   | 9780061122415 | 1988           |  
| 3       | Atomic Habits   | 9780735211292 | 2018           |  
| 4       | Python Programming | 9781234567890 | 2025           |  
+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM donation_history;
```

donation_id	book_id	donor_name	donation_date
1	4	Prem Kumar	2026-07-30

```
1 row in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_books_isbn
-> ON books(isbn);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
-> FROM books
-> WHERE isbn='9781234567890';
```

book_id	title	isbn	published_year
4	Python Programming	9781234567890	2025

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
3	Atomic Habits	9780735211292	2018
4	Python Programming	9781234567890	2025

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
3	Ravi Verma	ravi.verma@example.com

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM loans;
```

loan_id	member_id	book_id	loan_date
1	1	1	2026-07-01
2	3	3	2026-07-03
3	1	3	2026-07-05
4	3	1	2026-07-07

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM donation_history;
```

donation_id	book_id	donor_name	donation_date
1	4	Prem Kumar	2026-07-30

```
1 row in set (0.00 sec)
```